Namita Pokhrel

Pandas

March 22, 2026

In [24]:
```
import numpy as np
import pandas as pd
```

In [3]:
```
data = {'animal': ['cat', 'cat', 'snake', 'dog', 'dog', 'cat', 'snake', 'cat', 'dog
        'age': [2.5, 3, 0.5, np.nan, 5, 2, 4.5, np.nan, 7, 3],
        'visits': [1, 3, 2, 3, 2, 3, 1, 1, 2, 1],
        'priority': ['yes', 'yes', 'no', 'yes', 'no', 'no', 'no', 'yes', 'no', 'no'

labels = ['a', 'b', 'c', 'd', 'e', 'f', 'g', 'h', 'i', 'j']

df = pd.DataFrame(data, index=labels)
df
```

Out[3]:

	animal	age	visits	priority
a	cat	2.5	1	yes
b	cat	3.0	3	yes
c	snake	0.5	2	no
d	dog	NaN	3	yes
e	dog	5.0	2	no
f	cat	2.0	3	no
g	snake	4.5	1	no
h	cat	NaN	1	yes
i	dog	7.0	2	no
j	dog	3.0	1	no

In [4]:
```
df.info()
df.describe()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 10 entries, a to j
Data columns (total 4 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   animal    10 non-null     object
 1   age       8 non-null      float64
 2   visits    10 non-null     int64
 3   priority  10 non-null     object
dtypes: float64(1), int64(1), object(2)
memory usage: 400.0+ bytes
```

Out[4]:

	age	visits
count	8.000000	10.000000
mean	3.437500	1.900000
std	2.007797	0.875595
min	0.500000	1.000000
25%	2.375000	1.000000
50%	3.000000	2.000000
75%	4.625000	2.750000
max	7.000000	3.000000

In [5]:
```
df.head(3)
```

Out[5]:

	animal	age	visits	priority
a	cat	2.5	1	yes
b	cat	3.0	3	yes
c	snake	0.5	2	no

In [6]:
```
df[['animal', 'age']]
```

Out[6]:

	animal	age
a	cat	2.5
b	cat	3.0
c	snake	0.5
d	dog	NaN
e	dog	5.0
f	cat	2.0
g	snake	4.5
h	cat	NaN
i	dog	7.0
j	dog	3.0

In [7]:
```
df[df['visits'] > 2]
```

Out[7]:

	animal	age	visits	priority
b	cat	3.0	3	yes
d	dog	NaN	3	yes
f	cat	2.0	3	no

In [8]:
```
df[df['age'].isna()]
```

Out[8]:

	animal	age	visits	priority
d	dog	NaN	3	yes
h	cat	NaN	1	yes

In [9]:
```
df[(df['animal'] == 'cat') & (df['age'] < 3)]
```

Out[9]:

	animal	age	visits	priority
a	cat	2.5	1	yes
f	cat	2.0	3	no

In [10]:
```
df['visits'].sum()
```

Out[10]:
```
np.int64(19)
```

In [11]:
```
df.groupby('animal')['age'].mean()
```

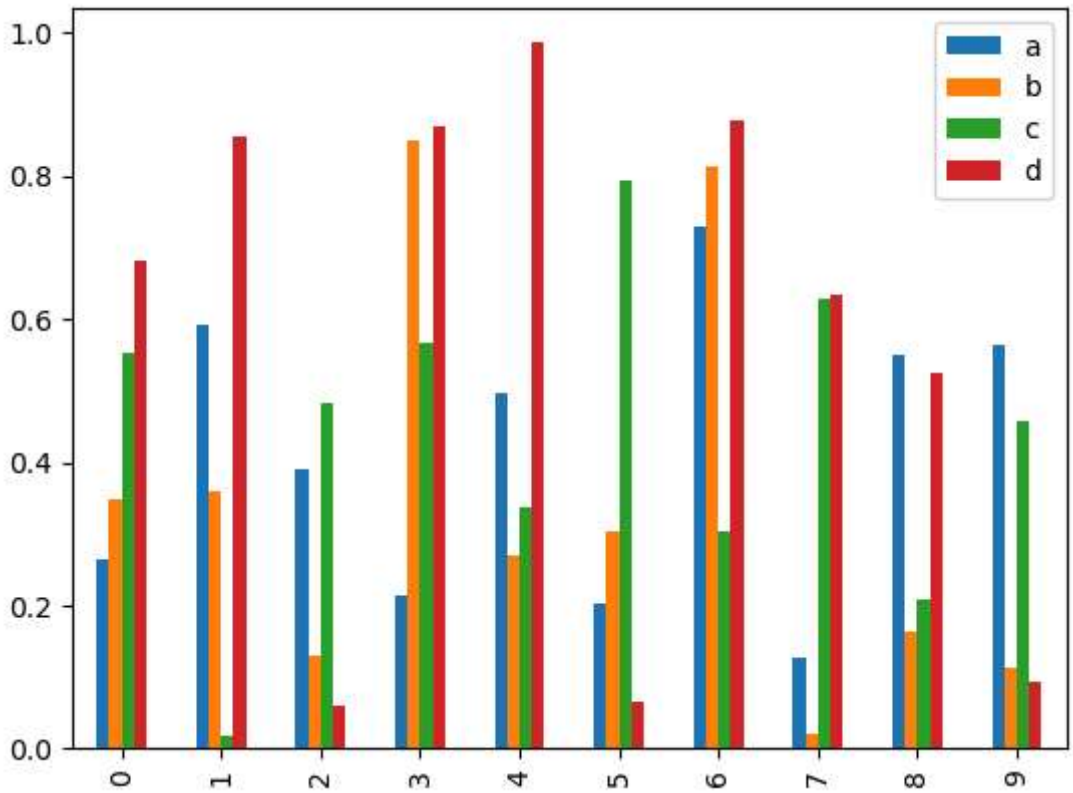
Out[11]:
```
animal
cat      2.5
dog      5.0
snake    2.5
Name: age, dtype: float64
```

In [12]:
```
df['animal'].value_counts()
```

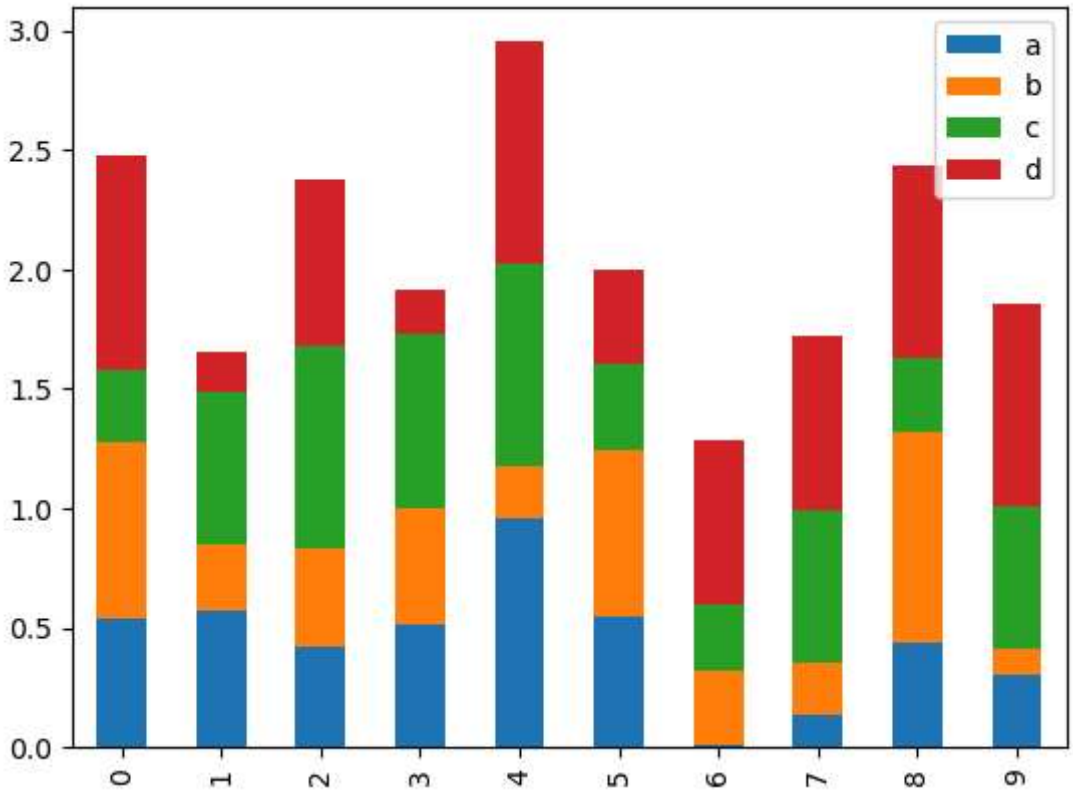
Out[12]:
```
animal
cat      4
dog      4
snake    2
Name: count, dtype: int64
```

In [13]:
```
import matplotlib.pyplot as plt
```

In [14]:
```
df = pd.DataFrame(np.random.rand(10,4),columns=['a','b','c','d'])
df.plot(kind='bar')
plt.show()
```

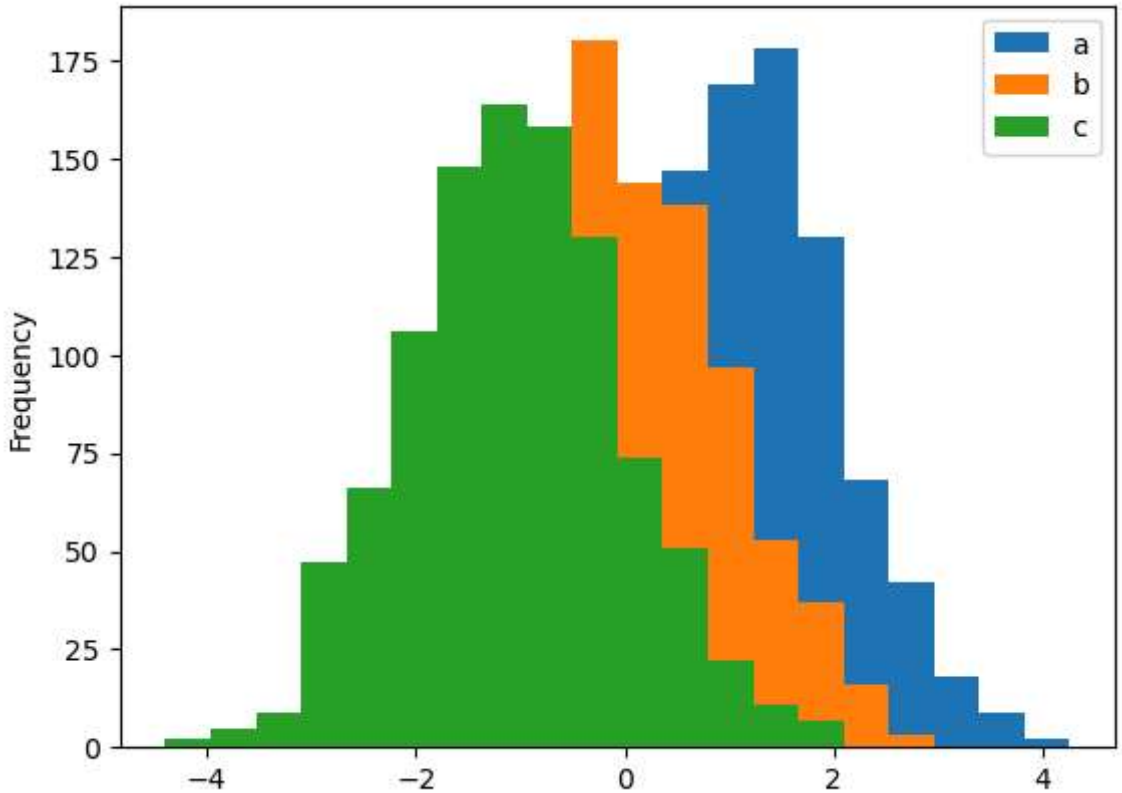


In [15]:
```
df = pd.DataFrame(np.random.rand(10,4),columns=['a','b','c','d'])
df.plot(kind='bar', stacked=True)
plt.show()
```



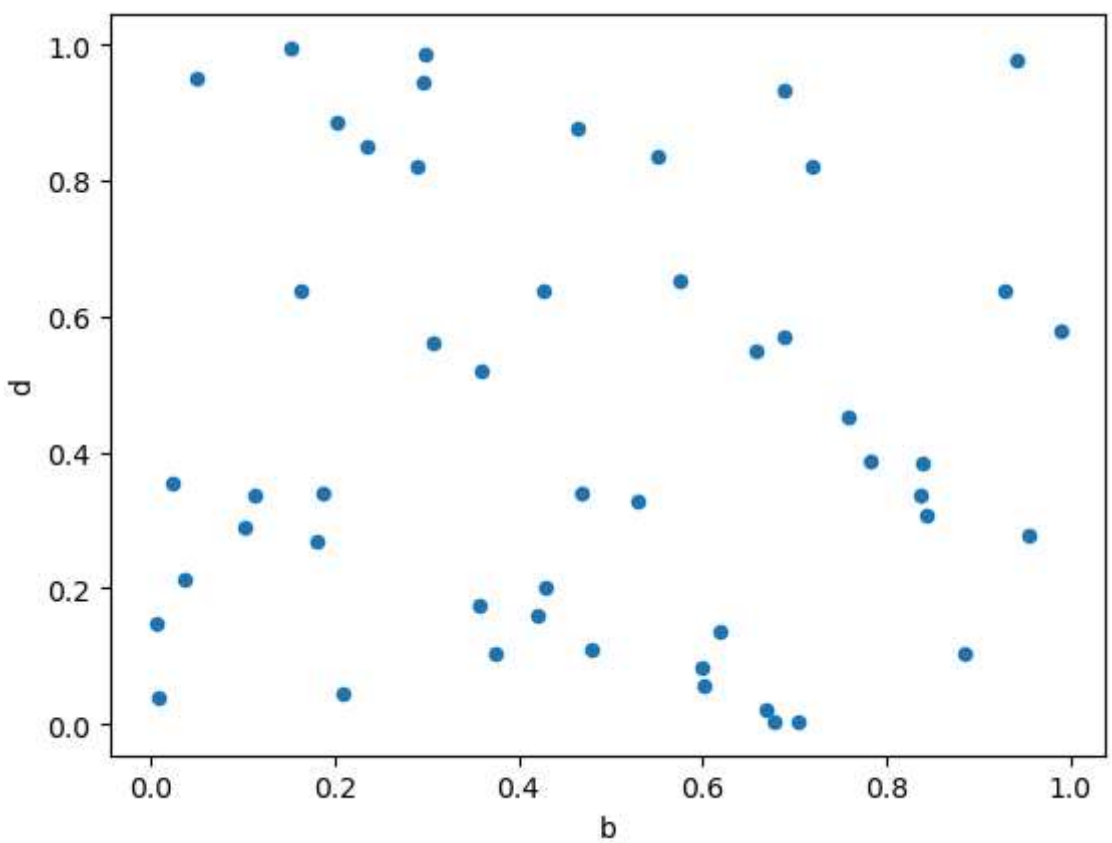
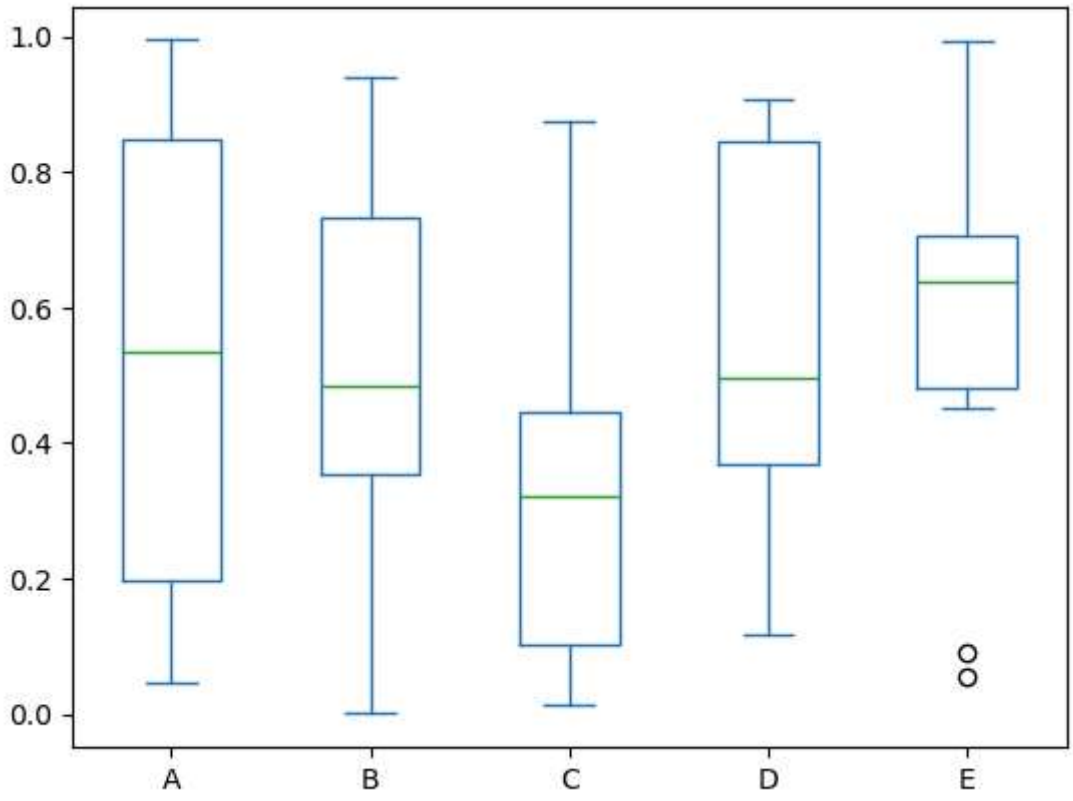
In [16]:
```
df = pd.DataFrame({'a':np.random.randn(1000)+1,
                   'b':np.random.randn(1000),
                   'c':np.random.randn(1000) - 1},
                   columns=['a', 'b', 'c'])

df.plot(kind='hist', bins=20)
plt.show()
```



In [17]:
```
df = pd.DataFrame(np.random.rand(50, 4), columns=['a', 'b', 'c', 'd'])

df.plot(kind='scatter', x='b', y='d')
plt.show()
```



In [18]:
```
df = pd.DataFrame(np.random.rand(10, 5), columns=['A', 'B', 'C', 'D', 'E'])

df.plot(kind='box')
plt.show()
```